

Training Dynamics of RNNs and GRUs

CS728: Programming Assignment 2

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1 Task 1: Memorization Task (A1–A5)

All five runs use `nhid=50`, `lr=0.01`, `min_length=50`, `max_length=200`. Per-run figures show: (1) the main panel (gradient norm histogram, saturation distance, training NLL), (2) the gradient-through-time histogram at four checkpoints, (3) gradient norm vs. iteration, and (4) $\rho(W_{hh})$.

1.1 A1: RNN tanh, No Clipping

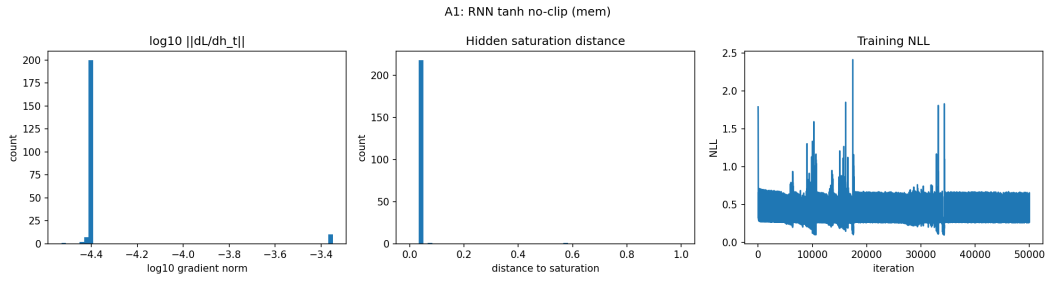


Figure 1: A1 – log10 gradient norm histogram (left), hidden saturation distance (center), training NLL (right).

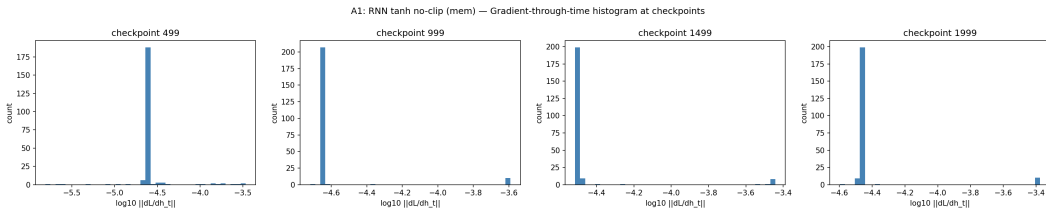


Figure 2: A1 – Gradient-through-time histogram at four training checkpoints.

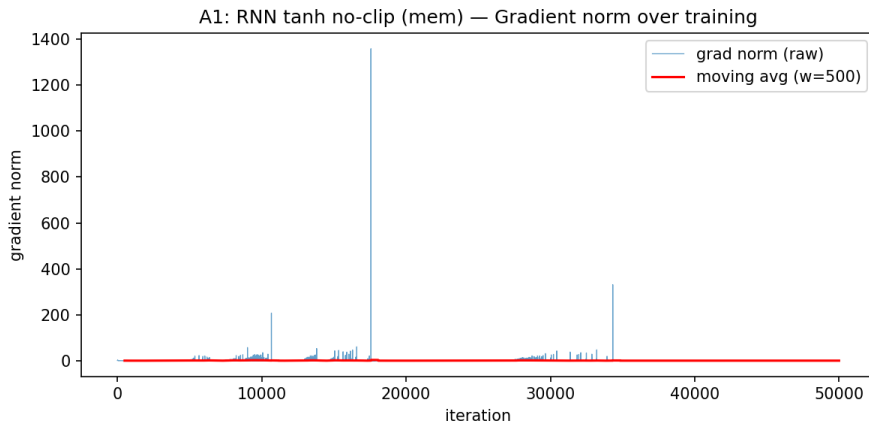


Figure 3: A1 – Gradient norm vs. iteration (blue: raw; red: moving average).

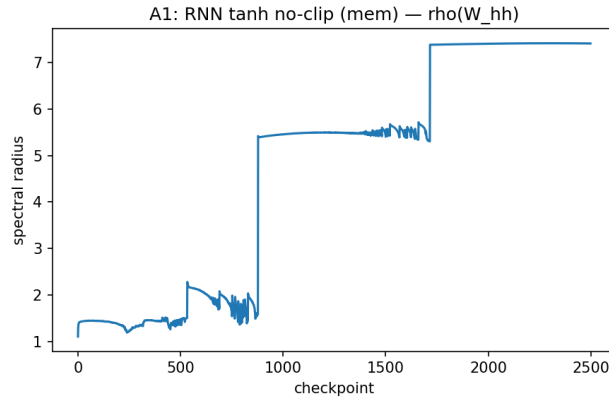


Figure 4: A1 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. The $\log_{10} \|\partial L / \partial h_t\|$ histogram shows almost all mass in a single spike at approximately -4.4 , with a tiny secondary bar near -3.4 . The distribution spans only about -4.86 to -3.4 , with the bulk of timesteps at the lower edge. This indicates severe vanishing gradients — the gradient signal decays to $\approx 10^{-4.4}$ by the time it reaches early timesteps.

Q2 – Saturation distance histogram. The hidden saturation distance histogram is sharply peaked near 0.05 , meaning most units are saturated ($|h| \approx 1$). Since $d(h) = 1 - |h|$, a distance near zero implies $h \approx \pm 1$, where the tanh derivative $1 - h^2 \approx 0$. Saturation is the primary driver of vanishing in A1.

Gradient-through-time histogram. Across all four checkpoints (499, 999, 1499, 1999) the distribution remains an almost identical spike near -4.4 – -4.5 with no spreading or shifting. This stationarity shows the gradient pathology is locked in from early training and never improves.

Gradient norm over training. The gradient norm is mostly near zero throughout 50 000 iterations, punctuated by rare sudden spikes: one reaching ≈ 1350 around iteration 18 000 and another ≈ 330 around iteration 34 000. The moving average stays essentially flat near zero. These isolated explosions correspond to the sudden jumps in $\rho(W_{hh})$ and confirm that without clipping, gradient explosions occur intermittently but produce no sustained learning.

Training NLL. The NLL starts at ≈ 1.8 , quickly drops and then fluctuates noisily between roughly 0.2 and 0.8 , with occasional upward spikes (co-located with the gradient explosions) reaching ≈ 2.4 . There is no sustained downward trend after the initial drop.

Q3 – Effect of clipping (comparison with A2/A3). Without clipping the gradient norm histogram is a single spike at -4.4 ; saturation distance is near 0 ; ρ jumps uncontrollably; and the gradient norm over training is flat with intermittent explosions. These features jointly prevent sustained learning. See A2 and A3 for the effects of different clipping thresholds.

Q5 – $\rho(W_{hh})$. ρ starts near 1.2 and undergoes three sudden step-like jumps to ≈ 2.2 , ≈ 5.4 , and ≈ 7.4 , where it remains flat. Each jump matches a gradient explosion spike in the norm-over-training plot. The very large final rho ($\gg 1$) and tanh saturation coexist at



different timesteps.



1.2 A2: RNN tanh, Moderate Clipping (cutoff = 0.05)

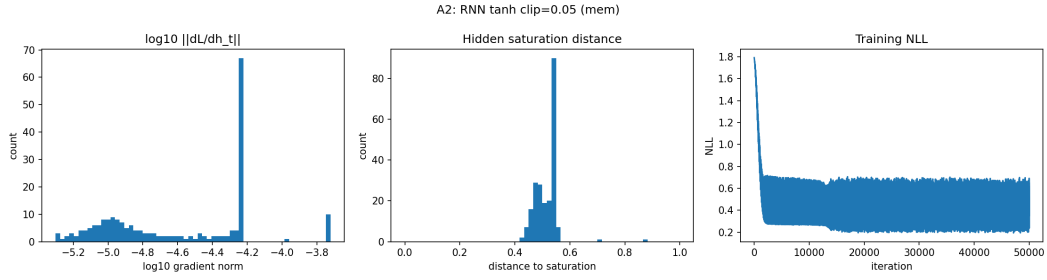


Figure 5: A2 – log10 gradient norm histogram (left), hidden saturation distance (center), training NLL (right).

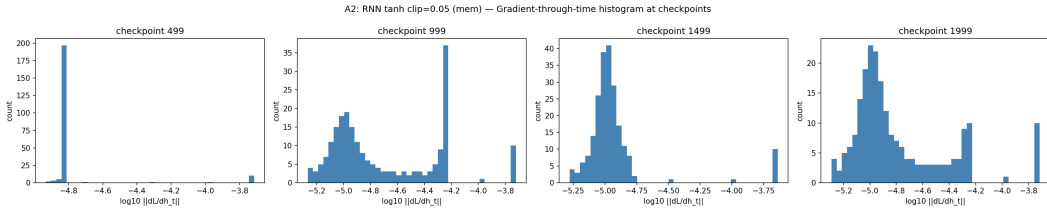


Figure 6: A2 – Gradient-through-time histogram at four training checkpoints.

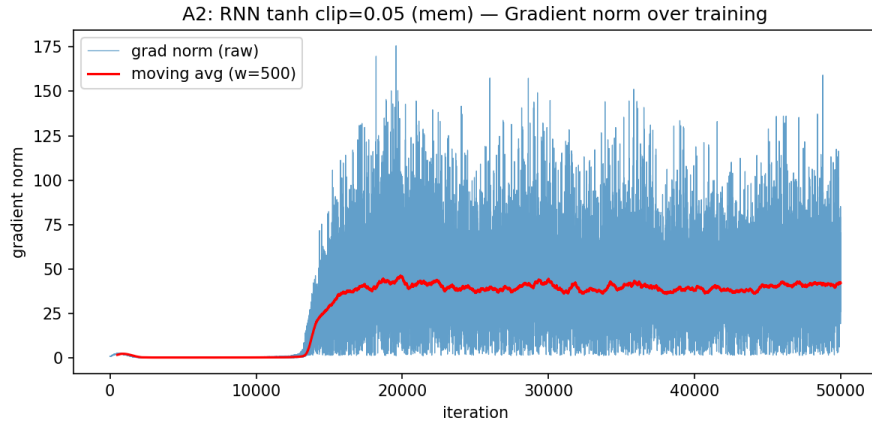


Figure 7: A2 – Gradient norm vs. iteration.

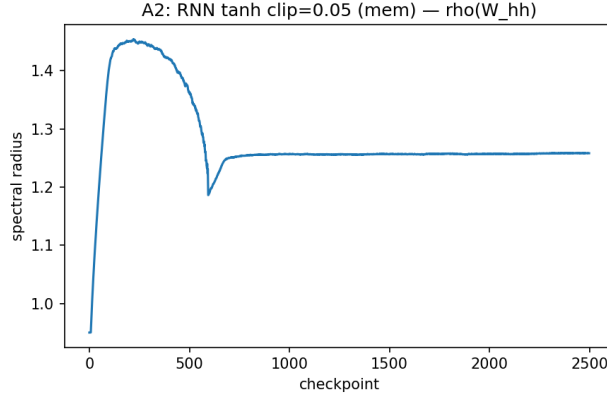


Figure 8: A2 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. With moderate clipping (cutoff = 0.05) the $\log_{10}$ gradient norm histogram is broadly spread from about -5.2 to -3.8 , with a peak near -4.2 . The distribution spans roughly 1.4 log-units — substantially wider than A1, indicating a wider variety of gradient magnitudes reaching different timesteps.

Q2 – Saturation distance histogram. The saturation distance is concentrated between 0.4 and 0.6, centered near 0.5 (the maximum, corresponding to $h = 0$). Hidden units are now in the unsaturated linear regime. Moderate clipping prevents the explosive weight growth that drove units to saturation in A1.

Gradient-through-time histogram. At checkpoint 499 the distribution is a narrow spike near -4.8 . By checkpoint 999 it has noticeably broadened and shifted, spreading from -5.2 to -3.8 with multiple humps. Checkpoints 1499 and 1999 maintain a similar multi-modal spread centered around -5.0 . This temporal evolution — from narrow to broad — is the key signature of improving gradient flow, a pattern absent in A1 and A3.

Gradient norm over training. The gradient norm is near zero for the first ~ 12000 iterations, then rises sharply to a sustained level of $\approx 30\text{--}175$ (raw), with the moving average stabilising around 40 from iteration ~ 15000 onwards. The clean phase transition from near-zero to high and sustained norms confirms that clipping gives the model time to build the right weight structure before gradients start flowing meaningfully.

Training NLL. The NLL shows a clear exponential-like decay from ≈ 1.8 down to a sustained floor around 0.2–0.7, reached after only a few thousand iterations. This is the strongest NLL reduction of all A-series runs, consistent with the best gradient flow.

Q3 – Effect of clipping. Comparing A1 vs A2: gradient norm shifts from spike at -4.4 to broad spread at -4.2 ; saturation distance shifts from near 0 to near 0.5; ρ changes from explosive jumps to a controlled stable value; gradient norm over training changes from flat-with-spikes to a clean rise-and-plateau; gradient-through-time evolves in A2 but is stationary in A1. Moderate clipping is unambiguously beneficial across all diagnostics.

Q5 – $\rho(W_{hh})$. Rho rises from ≈ 0.97 to a peak of ≈ 1.44 (checkpoint ~ 200), then gradually decreases to ≈ 1.19 , and stabilises around 1.25 for the remainder of training.



1.3 A3: RNN tanh, Aggressive Clipping (cutoff = 0.01)

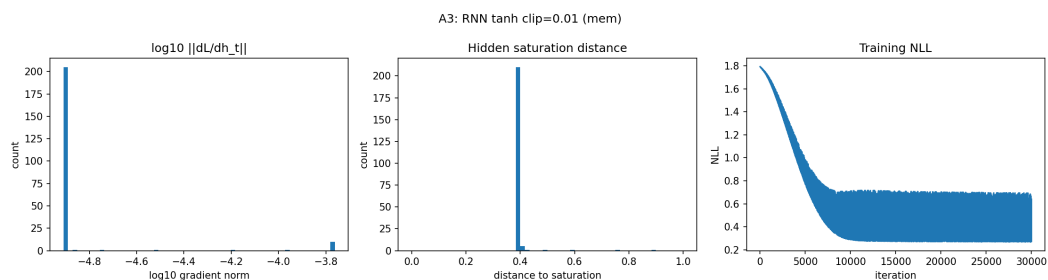


Figure 9: A3 – log10 gradient norm histogram (left), hidden saturation distance (center), training NLL (right).

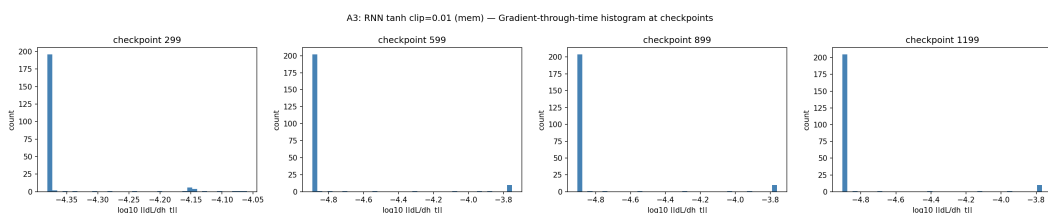


Figure 10: A3 – Gradient-through-time histogram at four training checkpoints.

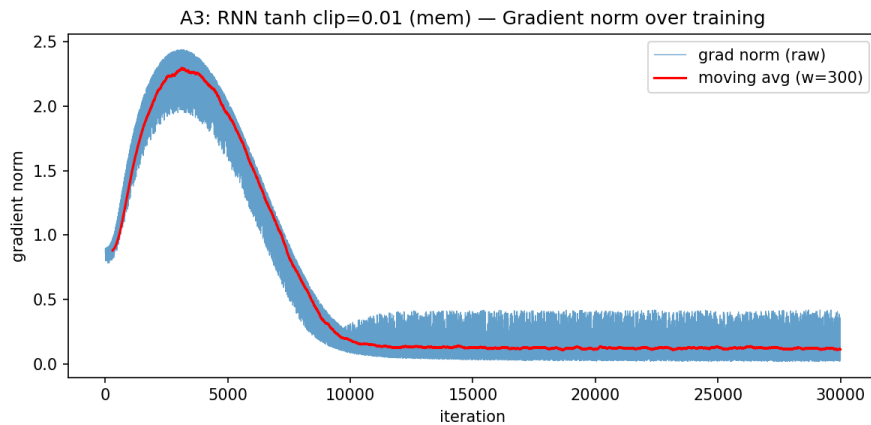


Figure 11: A3 – Gradient norm vs. iteration.

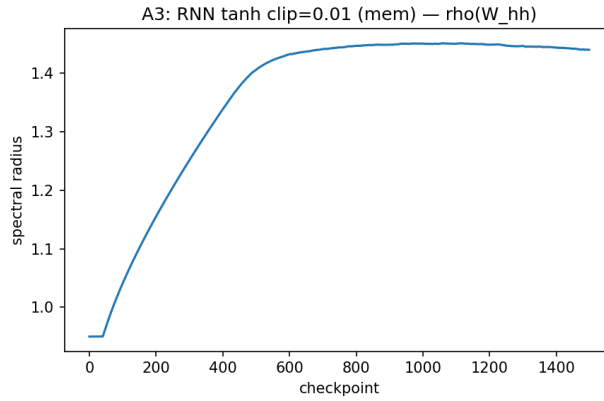


Figure 12: A3 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. With aggressive clipping (cutoff = 0.01), the gradient norm histogram is a very narrow spike at approximately -4.8 , more negative than A1 (-4.4). Aggressive clipping paradoxically produces worse gradient flow than no clipping: updates are so small that weights cannot grow enough to amplify gradients.

Q2 – Saturation distance histogram. The saturation distance is sharply peaked at ≈ 0.4 , slightly more unsaturated than A1 (near 0) but more saturated than A2 (near 0.5). Aggressive clipping prevents extreme saturation but cannot reach A2’s clean unsaturated regime.

Gradient-through-time histogram. All four checkpoints (299, 599, 899, 1199) show almost identical narrow spikes near -4.8 with a tiny secondary bar near -3.8 . There is virtually no evolution, confirming the gradient pathology is fixed early and never changes — in strong contrast to A2.

Gradient norm over training. The norm rises smoothly from ≈ 0.9 to a peak of ≈ 2.4 around iteration 3000–4000, then decays gradually to a floor near 0.1–0.4 by iteration ~ 9000 , remaining there for the rest of training. This bell-shaped profile is qualitatively different from both A1 (flat near zero with spikes) and A2 (rise to sustained plateau), and suggests the model settles into a low-gradient, suboptimal regime.

Training NLL. The NLL shows a clean smooth decay from ≈ 1.8 to a floor around 0.3–0.8. The shape is similar to A2 but the floor is slightly higher and training terminates earlier ($\sim 30\,000$ iterations), suggesting convergence to a suboptimal solution.

Q3 – Effect of clipping. Across A1, A2, A3: gradient norm at -4.4 (A1), spread at -4.2 (A2, best), spike at -4.8 (A3, most vanished); saturation at ≈ 0 (A1), ≈ 0.5 (A2), ≈ 0.4 (A3); gradient-through-time stationary in A1 and A3 but evolving in A2; gradient norm over training with flat-spikes (A1), rise-plateau (A2), bell-decay (A3). Moderate clipping (A2) remains the best-performing strategy.

Q5 – $\rho(W_{hh})$. The rho curve for A3 is a clean monotonic rise from ≈ 0.95 to ≈ 1.44 , plateauing and remaining there for the rest of training. The stable plateau at 1.44 combined with the narrow gradient spike at -4.8 confirms the model is stuck.



1.4 A4: GRU, No Clipping

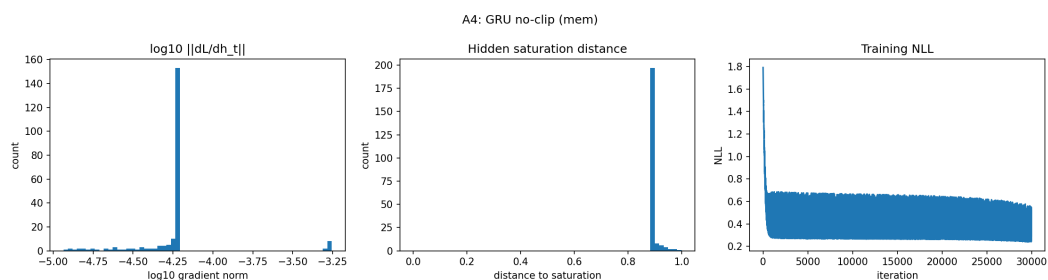


Figure 13: A4 – log10 gradient norm histogram (left), hidden saturation distance (center), training NLL (right).

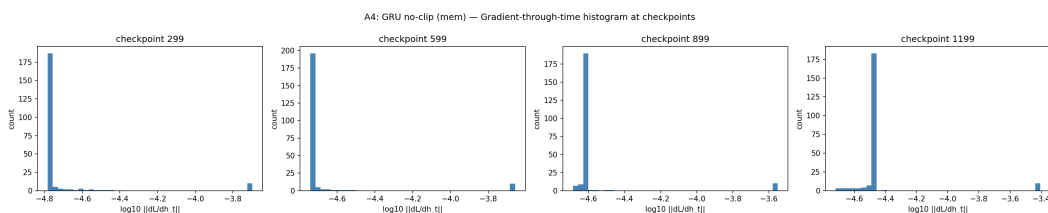


Figure 14: A4 – Gradient-through-time histogram at four training checkpoints.

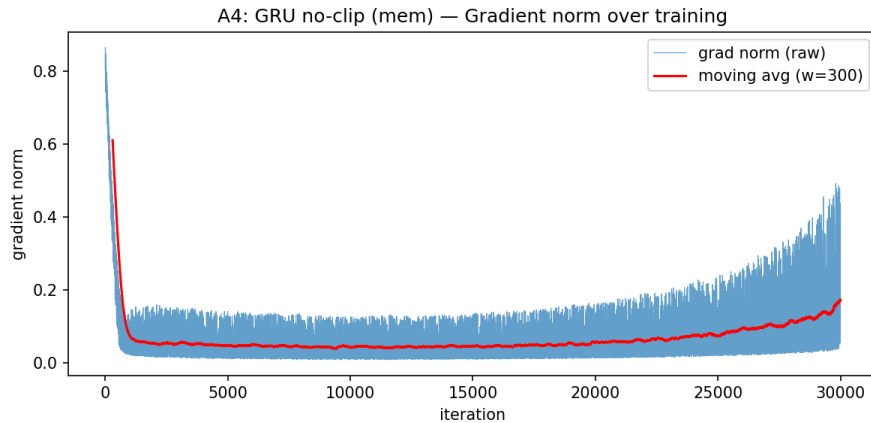


Figure 15: A4 – Gradient norm vs. iteration.

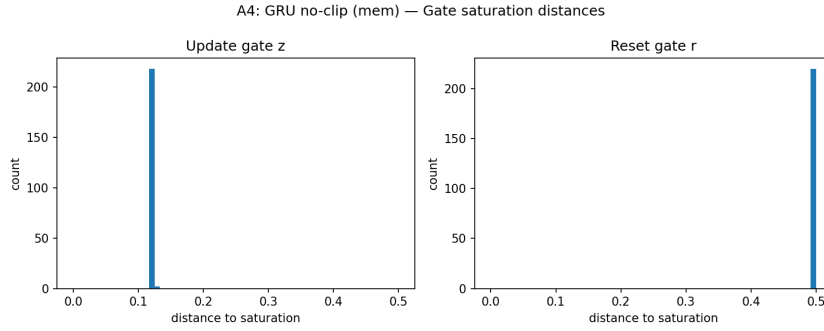


Figure 16: A4 – Gate saturation distances: update gate z (left), reset gate r (right).

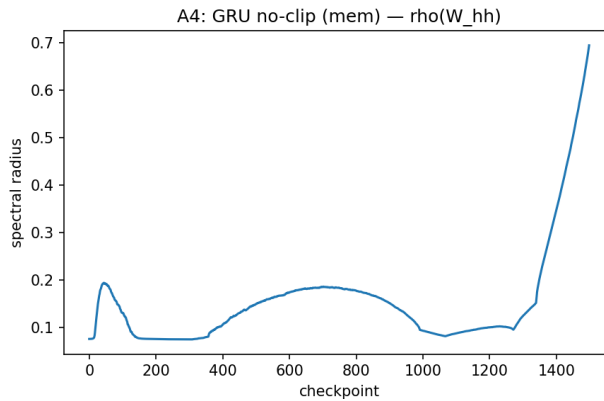


Figure 17: A4 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. The GRU without clipping shows a gradient norm distribution with a dominant spike near -4.25 and noticeable spread from -5.0 to -3.25 , with a small secondary bar near -3.25 . Compared to the RNN runs, the histogram is slightly wider and less negative, indicating marginally better gradient propagation from the gating mechanism.

Q2 – Saturation distance histogram. The hidden saturation distance shows a sharp spike near 0.9 , meaning candidate $\tanh$ hidden states are almost completely unsaturated ($\tilde{h} \approx 0$). The GRU interpolation

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

combined with the saturated update gate (see Q4) keeps hidden states near h_{t-1} , driving candidate activations toward zero.

Gradient-through-time histogram. All four checkpoints (299, 599, 899, 1199) show a narrow spike near -4.6 – -4.8 with a small secondary bar near -3.8 . The distribution barely evolves across checkpoints — the GRU without clipping does not progressively improve its through-time gradient flow, despite the architectural advantages of gating.

Gradient norm over training. The norm starts high (≈ 0.86), drops sharply within the first ~ 1000 iterations to a floor around 0.05 – 0.15 , then very gradually increases after



iteration $\sim 20\,000$. The moving average trend turns upward after checkpoint ~ 1500 , reaching ≈ 0.17 by the end of training, aligning with the late spike in $\rho(W_{hh})$. This profile is qualitatively different from all three RNN runs: it starts high rather than building up.

Training NLL. The NLL starts at ≈ 1.8 , drops rapidly within a few hundred iterations to a floor of ≈ 0.25 – 0.65 , and fluctuates without a clear downward trend for the remaining $\sim 25\,000$ iterations. The initial drop is slightly faster than A2/A3 but the final floor is similar.

Q4 – RNN vs GRU gate saturation. The update gate z distribution is spiked at ≈ 0.12 , meaning z values are near 0 or 1 (saturated; $d(v) = \min(v, 1 - v) \approx 0.12$ implies $v \approx 0.12$ or $v \approx 0.88$). The reset gate r is spiked at ≈ 0.5 — completely unsaturated. The saturated update gate enforces near-binary copy-or-replace decisions at every timestep.

Q5 – $\rho(W_{hh})$. Rho stays very low (0.08–0.2) for most of training with gentle oscillations, then spikes sharply to ≈ 0.70 in the final checkpoints — a sign of emerging instability without clipping. Even at its peak, the GRU rho is far below the RNN values (1.2–7.4).



1.5 A5: GRU, Moderate Clipping (cutoff = 0.05)

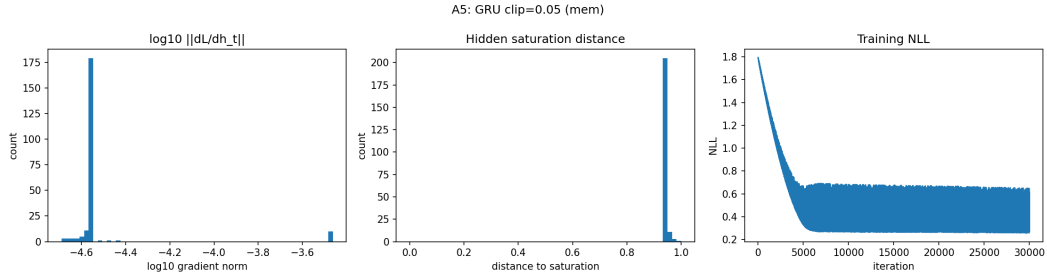


Figure 18: A5 – log10 gradient norm histogram (left), hidden saturation distance (center), training NLL (right).

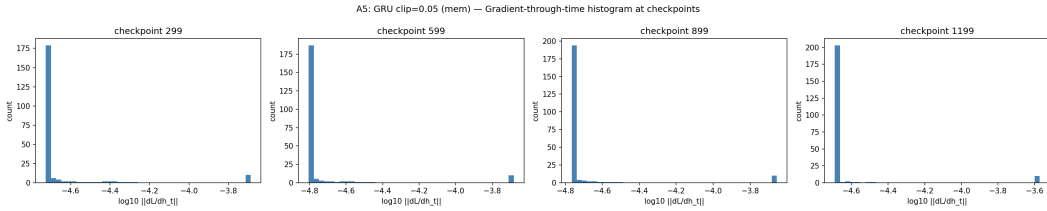


Figure 19: A5 – Gradient-through-time histogram at four training checkpoints.

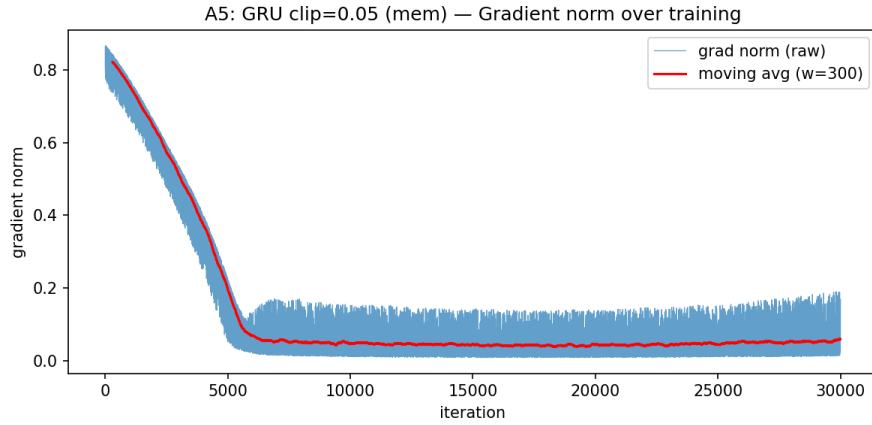


Figure 20: A5 – Gradient norm vs. iteration.

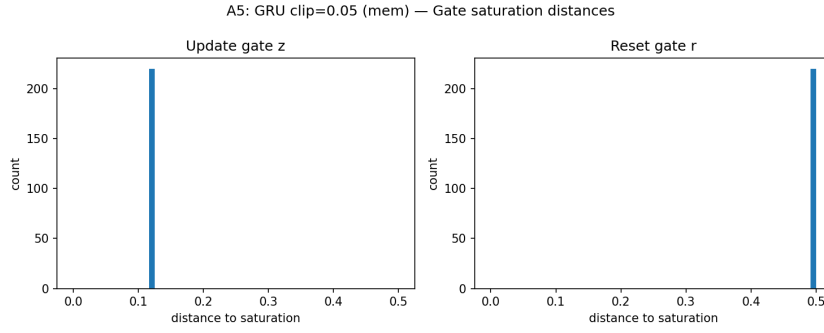


Figure 21: A5 – Gate saturation distances: update gate z (left), reset gate r (right).

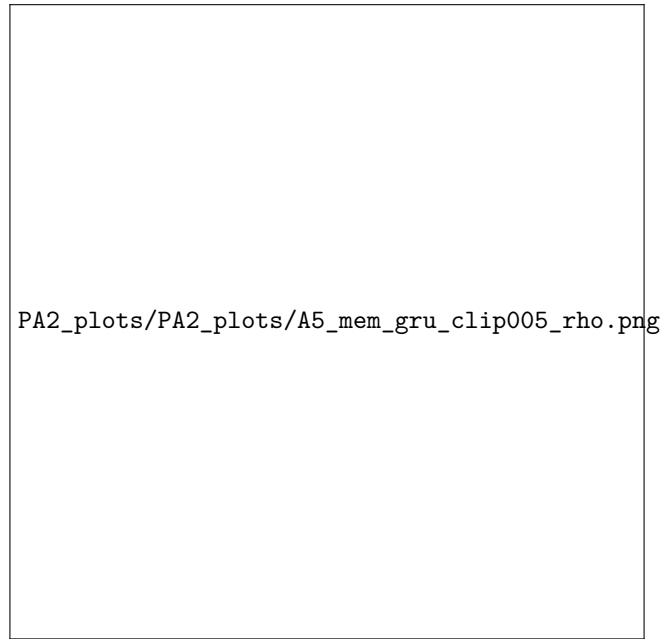


Figure 22: A5 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. With clipping at 0.05, the GRU gradient norm histogram is a very narrow spike at approximately -4.6 , with a small secondary bar near -3.6 . The peak is slightly more negative and narrower than A4 (-4.25 , broader), mirroring the A3 vs A1 pattern: clipping slightly worsens the gradient norm distribution for GRUs.

Q2 – Saturation distance histogram. The saturation distance is a sharp spike near 0.95 – 1.0 , even closer to full unsaturation than A4 (≈ 0.9). Clipping further restricts weight updates, keeping candidate hidden states even closer to zero.

Gradient-through-time histogram. All four checkpoints (299, 599, 899, 1199) again show nearly identical narrow spikes near -4.6 – -4.8 , with a tiny secondary bar near -3.8 . As with A4, the distribution is stationary. Adding clipping to the GRU does not improve through-time gradient flow.

Gradient norm over training. The norm starts at ≈ 0.86 , drops smoothly and steeply to a floor around 0.05 – 0.10 within the first ~ 5000 iterations, and then remains flat



throughout. Comparing to A4: clipping eliminates the late-stage norm growth. The moving average is nearly horizontal from iteration ~ 6000 onwards.

Training NLL. Nearly identical to A4: a rapid initial drop from ≈ 1.8 to a floor of $\approx 0.25\text{--}0.65$, with no further trend. Clipped and unclipped GRUs achieve essentially the same training NLL on the mem task.

Q3 – Effect of clipping on GRU. Comparing A4 vs A5: gradient norm shifts from -4.25 (broader) to -4.6 (narrower); saturation distance from ≈ 0.9 to ≈ 0.95 ; gradient norm over training changes from a gradual upward trend (A4) to a flat plateau (A5); rho changes from irregular oscillation with a late spike (A4) to a two-arch pattern without a late spike (A5); NLL curves are indistinguishable. Clipping provides mild stability (preventing the late rho spike) but does not improve gradient flow or NLL for GRUs.

Q4 – Gate saturation (A5 vs A4). Gate plots for A5 are nearly identical to A4: update gate z at ≈ 0.12 (saturated) and reset gate r at ≈ 0.5 (unsaturated). Gate saturation behaviour is unaffected by gradient clipping.

Q5 – $\rho(W_{hh})$. Rho shows two distinct arching patterns: a rise to ≈ 0.175 (checkpoint ~ 280), a drop to ≈ 0.08 , a second rise to ≈ 0.188 (checkpoint ~ 1100), then a decline back to ≈ 0.085 . The two-arch oscillation and the absence of the late uncontrolled spike seen in A4 show that clipping stabilises weight dynamics. All values remain well below 1.



2 Task 2: Multiplication Task (B1–B2)

The multiplication task uses `classifType=lastLinear` — regression on the last hidden state to predict the product of two marked values. The model must retain information about two specific positions across the full variable-length sequence.

2.1 B1: RNN tanh, No Clipping (mul)

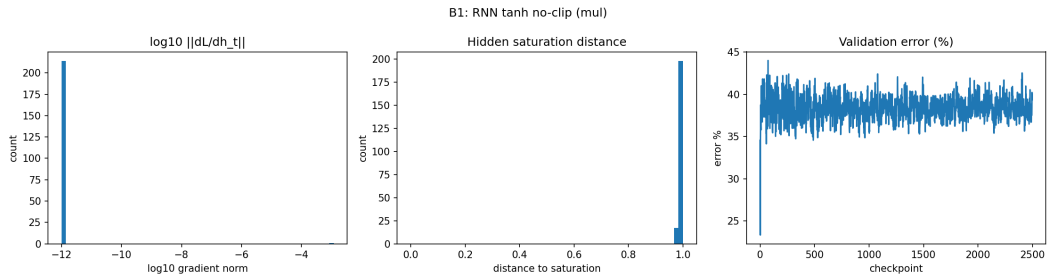


Figure 23: B1 – log10 gradient norm histogram (left), hidden saturation distance (center), validation error % (right).

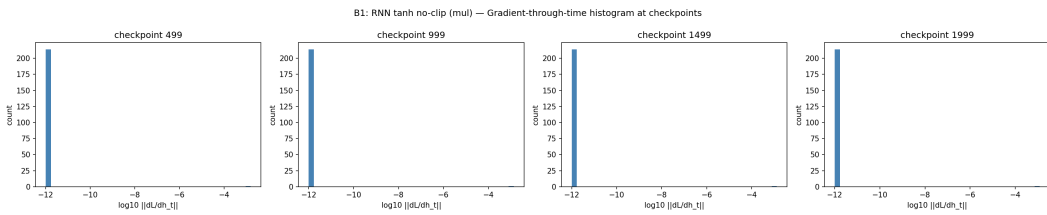


Figure 24: B1 – Gradient-through-time histogram at four checkpoints.

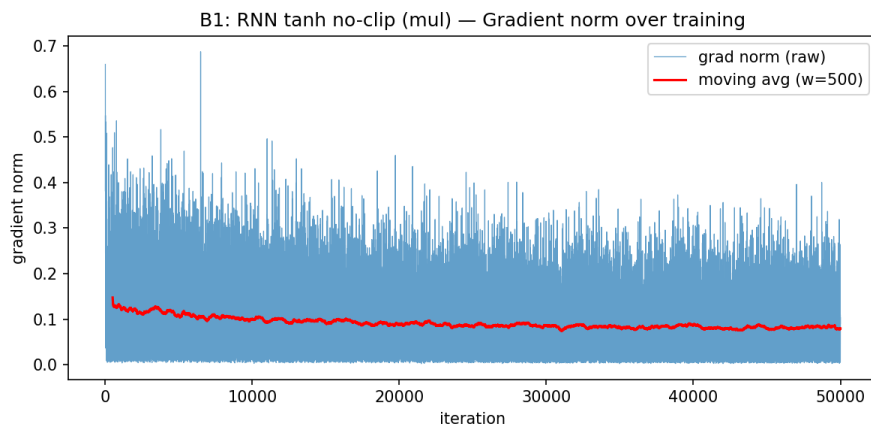


Figure 25: B1 – Gradient norm vs. iteration.

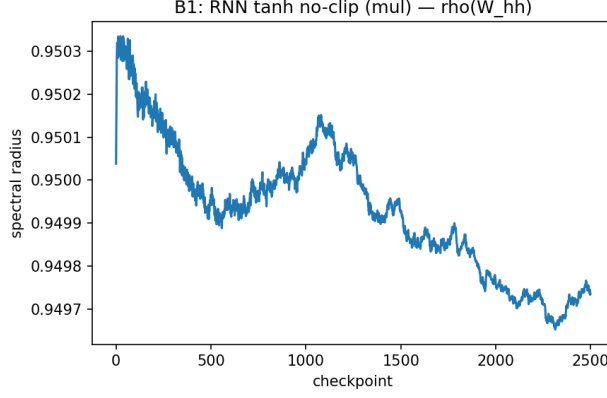


Figure 26: B1 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. The $\log_{10}$ gradient norm histogram shows a single extreme spike at -12 , with all mass concentrated there. This is dramatically more vanished than any mem-task run (around -4 to -5). At 10^{-12} , the gradient signal is numerically zero.

Q2 – Saturation distance histogram. The saturation distance is sharply spiked near 1.0. Hidden units are almost perfectly *unsaturated* ($h \approx 0$). This is a critical distinction from the mem task: the RNN is not saturated, yet gradients have completely vanished. The culprit is spectral decay: with $\rho(W_{hh}) \approx 0.95$ and sequence lengths of 50–200,

$$\rho(W_{hh})^T \approx 0.95^{100} \approx 0.006.$$

Saturation is *not* the cause here.

Gradient-through-time histogram. All four checkpoints (499, 999, 1499, 1999) show an identical spike at -12 . No evolution, no spreading. The gradient pathology is completely fixed from the start and never changes across 50 000 iterations.

Gradient norm over training. The gradient norm fluctuates around 0.05–0.7 throughout training. The moving average decays very gently from ≈ 0.14 to ≈ 0.08 . The norm is non-zero (unlike the -12 in the through-time histogram) because the global gradient norm aggregates all parameters including output weights, not just $\partial L / \partial h_t$.

Validation error. Error fluctuates between approximately 35% and 43% with no systematic decreasing trend. The initial checkpoint shows a brief dip to $\approx 24\%$ before settling at the plateau. The model makes no meaningful progress on the multiplication task.

Q5 – $\rho(W_{hh})$. Rho fluctuates only in the range 0.9497 to 0.9503 throughout all ~ 2500 checkpoints — essentially constant. This near-constant value just below 1.0 directly causes the exponential gradient decay.



2.2 B2: GRU, No Clipping (mul)



Figure 27: B2 – log10 gradient norm histogram (left), hidden saturation distance (center), validation error % (right).

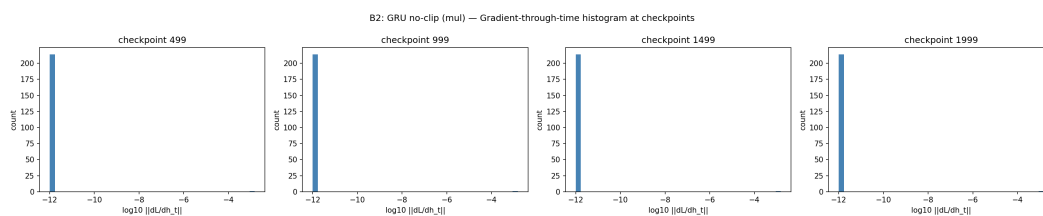


Figure 28: B2 – Gradient-through-time histogram at four checkpoints.

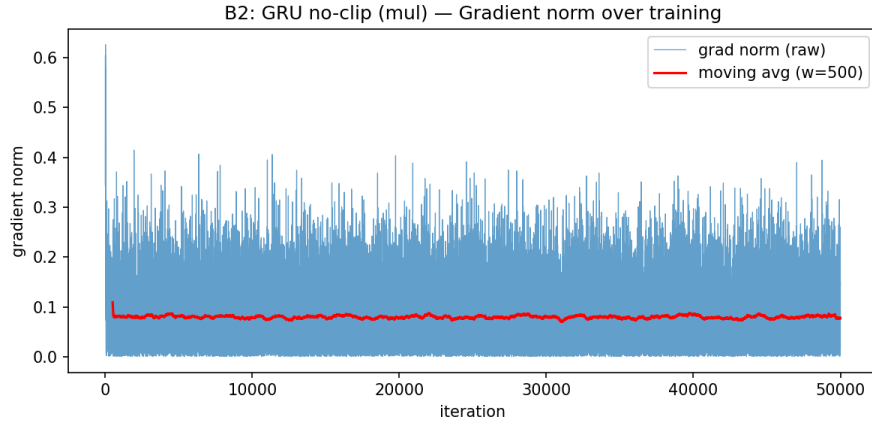
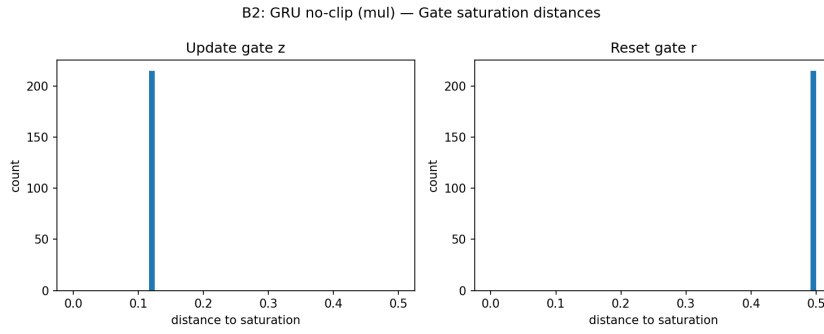
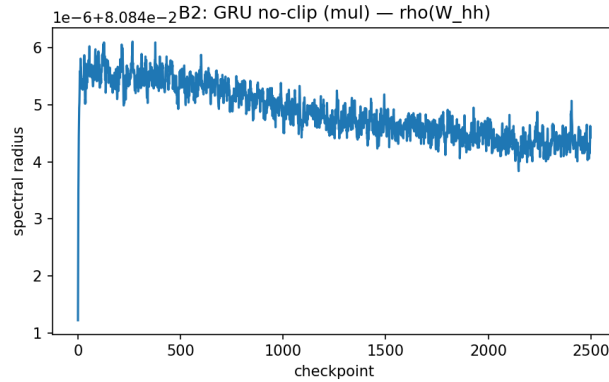


Figure 29: B2 – Gradient norm vs. iteration.

Figure 30: B2 – Gate saturation distances: update gate z (left), reset gate r (right).Figure 31: B2 – Spectral radius $\rho(W_{hh})$ over training checkpoints.

Q1 – Gradient norm histogram. The gradient norm histogram for B2 shows a spike at -12 , identical to B1. The GRU does not achieve better through-time gradient propagation than the RNN on this task.

Q2 – Saturation distance histogram. The saturation distance is spiked near 1.0,



identical to B1. Gradient vanishing in both mul-task runs is caused by spectral decay, not saturation.

Gradient-through-time histogram. All four checkpoints (499, 999, 1499, 1999) show an identical spike at -12 , mirroring B1 exactly. The GRU gating provides no relief from the vanishing gradient problem on this long-range dependency task.

Gradient norm over training. The gradient norm fluctuates around 0.05–0.4 throughout training, with a single early spike to ≈ 0.62 . The moving average is nearly flat at ≈ 0.08 –0.10 across all 50 000 iterations. The profile is very similar to B1.

Validation error. Error fluctuates between approximately 35% and 42%, with an initial dip to $\approx 25\%$ in the earliest checkpoints before settling into the plateau. Nearly identical to B1.

Q4 – RNN vs GRU on mul task. Comparing B1 and B2: gradient norm histograms are identical (spike at -12); saturation distances are identical (near 1.0); gradient-through-time histograms are stationary spikes at -12 for both; gradient norm over training traces are similar; both fail to learn. The GRU’s gating provides no advantage for the multiplication task within the training budget. The update gate z is saturated (distance ≈ 0.12) and the reset gate r is unsaturated (distance ≈ 0.50), the same regime as in A4/A5 — gate operating points are independent of the task.

Q5 – $\rho(W_{hh})$. The rho for B2 starts at ≈ 1.2 , rises to ≈ 5.9 within the first ~ 100 checkpoints, then gradually and noisily decays from ≈ 5.5 down to ≈ 4.1 over the remaining ~ 2400 checkpoints. The y-axis offset is 8.084×10^{-2} , so actual values are $\approx 0.08 + \text{displayed}$. Despite the high and changing rho, gradients remain pinned at -12 throughout — confirming that the GRU’s gate-modulated effective recurrence means the raw candidate W_{hh} spectral radius is not the sole determinant of gradient flow.



3 Summary

3.1 Q3: No Clipping vs. Clipping (RNN, mem task)

Comparing A1 (no clip), A2 (clip = 0.05), A3 (clip = 0.01) across all diagnostics: the gradient norm histogram changes from a single spike at -4.4 (A1) to a broad spread centered at -4.2 (A2, best) to a narrow spike at -4.8 (A3, worst). Saturation distance shifts from near 0 (A1) to near 0.5 (A2) to ≈ 0.4 (A3). The gradient-through-time histograms are stationary in A1 and A3 but show clear temporal broadening and shifting in A2 — the only run where gradient flow genuinely improves over training. The gradient norm over training is flat-with-explosions (A1), rise-to-sustained-plateau (A2), and bell-curve-declining-to-floor (A3). The $\rho(W_{hh})$ is explosive-with-jumps (A1), smoothly controlled and stabilising at 1.25 (A2), and monotone-rise-to-plateau-at-1.44 (A3). Training NLL tells the same story: A2 achieves the lowest sustained NLL floor while A1 is noisier and A3 converges to a slightly higher floor.

The conclusion is clear: moderate clipping (A2) provides the best balance, preventing explosive updates (A1) while still allowing sufficient weight growth (unlike A3 which over-restricts learning). The temporal evolution of the through-time histogram in A2 is the defining diagnostic.

3.2 Q4: RNN vs. GRU

Across gradient norm histograms, the GRU (A4, centered at -4.25 with visible spread) shows marginally better gradient propagation than RNN no-clip (A1, spike at -4.4) and is comparable to the clipped RNN runs. However, the gradient-through-time histograms tell a different story: both A4 and A5 have stationary, narrow spikes just like A1 and A3. None of the GRU runs show the temporal evolution seen in A2, meaning the GRU's improvement in the summary histogram does not translate to actual improvement in through-time gradient flow over training.

The hidden saturation distance in GRUs (≈ 0.9) is much higher than in RNNs (0 to 0.5), because the interpolation mechanism naturally keeps hidden states and candidate activations near zero. The gate saturation is identical across all GRU runs: update gate saturated at ≈ 0.12 , reset gate unsaturated at ≈ 0.5 .

The gradient norm over training has a fundamentally different shape for GRUs: it starts high and decays, whereas RNNs start low and build up (or stay near zero). The $\rho(W_{hh})$ for GRUs (≈ 0.08 – 0.7) is far lower than for RNNs (1.2–7.4), confirming that gating provides internal stability without requiring a high spectral radius.

On the mul task (B1 vs B2): all diagnostics are essentially identical. Both models have gradient norms pinned at -12 through time, unsaturated hidden states, flat gradient norm traces, and validation errors plateauing around 35–42%. The GRU provides no advantage over the RNN on this long-range dependency task within the given training budget.